Tail Annealing for Heavy-Tailed Flow Matching

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Abstract

Standard generative models struggle with heavy-tailed data: Lipschitz architectures cannot produce power-law tails from Gaussian noise, and interpolating between heavy-tailed data and Gaussians is ill-posed. We propose a simple fix: apply the soft-log transform $\phi(x) = \mathrm{sign}(x) \cdot \log(1 + |x|)$ coordinate-wise to data before training, then exponentiate samples after generation. A Hill diagnostic decides per-coordinate whether to transform, leaving light-tailed margins untouched at no added complexity. This compresses heavy tails into a range where standard flow matching succeeds, without heavy-tailed base distributions or architectural modifications. We provide theoretical intuition for why this works: the log-transform maps Pareto tails to exponentials, and the induced dynamics implement a form of tail annealing via power transformations. On a 144-configuration multivariate benchmark (3 copulas, d up to 100, 4 tail indices), Log-FM dominates specialized baselines on W_1, CVaR$_{99}$, and extreme-quantile metrics, and is the only method with zero severe divergences across 2,880 runs.

1. Introduction

Heavy-tailed distributions are ubiquitous in real-world phenomena: financial returns, insurance claims, earthquake magnitudes, network degrees, and climate extremes. In these domains, rare events dominate outcomes: asset returns exhibit empirically documented fat tails that Gaussian models systematically underestimate (Cont, 2001), and similar departures from Gaussianity are well-known in insurance (Embrechts et al., 2013) and natural-hazard data. Generating realistic synthetic data for stress testing, simulation, or data augmentation requires generative models that capture tail behavior.

Yet standard generative models fail on heavy-tailed data, both in theory and in practice. Jaini et al. (2020) proved that Lipschitz transformations preserve tail type: light-tailed inputs yield light-tailed outputs. Since most practical architectures (MLPs in GANs, coupling layers in normalizing flows, and diffusion denoisers) use Lipschitz-continuous functions for training stability, they cannot generate heavy tails from Gaussian noise without explicit tail-modifying components.

Prior approaches. Existing methods address heavy tails through two strategies. Jaini et al. (2020) replace Gaussian noise with heavy-tailed base distributions (Student-t), but this introduces training instability. Hickling & Prangle (2025) take a different approach: generate samples from a standard flow, then apply a tail-modifying transformation to the *output*. Their method requires estimating the tail parameter $\lambda = 1/\nu$ via the Hill estimator.

Our approach. We propose a simple fix: transform data via $\phi(x) = \mathrm{sign}(x) \cdot \log(1 + |x|)$ before training, run standard flow matching in log-space, then apply ϕ^{-1} to generated samples. Unlike Hickling & Prangle (2025) who transform outputs with estimated tail parameters, our input transform is parameter-free: the transformation ϕ works regardless of the true tail index.

We provide theoretical intuition for why this works. The log-transform compresses heavy tails: Pareto becomes approximately exponential. In log-space, both data and Gaussian noise have light tails, so standard interpolation is well-posed. The induced dynamics in original space implement a form of tail annealing via power transformations $X_0^{\alpha_t}$.

Contributions.

- We show that a parameter-free log-transform makes standard flow matching work for heavy-tailed data, without requiring tail estimation or architectural changes (§4).
- We analyze the tail annealing mechanism: the log-transform maps heavy tails to exponential tails, and the induced dynamics implement power transformations $X_0^{\alpha_t}$ that continuously adjust the tail index (§3).
- Our 144-configuration multivariate benchmark (3 copulas, 4 dimensions up to $d = 100$, 4 tail indices, 20

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replications, two margin types) shows Log-FM dominates baselines on tail metrics and is the only method with zero severe divergences (§5). Real-data validation on Fama–French is in Appendix E.2.

2. Background

We review heavy-tailed distributions and the fundamental barriers they pose for generative modeling.

2.1. Heavy-Tailed Distributions

A distribution is *heavy-tailed* if its tails decay slower than any exponential, formally $\mathbb{E}[e^{\lambda X}] = \infty$ for all $\lambda > 0$ (Nair et al., 2022). Heavy-tailed distributions arise naturally in finance (Cont, 2001), insurance, climate science, and network traffic, where rare events have outsized impact.

The canonical example is the Pareto distribution with survival function $\mathbb{P}(X > t) = t^{-1/\gamma}$ for $t \geq 1$, where $\gamma > 0$ is the shape parameter (the GPD shape, equivalently the extreme-value index). Larger γ means heavier tails: $\gamma < 1$ for finite mean, $\gamma < 1/2$ for finite variance. The Student-t distribution with ν degrees of freedom has Pareto-like tails with effective $\gamma = 1/\nu$. Lognormal has $\gamma = 0$ (Gumbel max-domain of attraction): heavy in the MGF sense, but with no polynomial decay rate.

The log-transform. A key observation: the logarithm maps heavy tails to lighter tails. If $X \sim \text{Pareto}(\gamma)$, then $\log X \sim \text{Exp}(1/\gamma)$, an exponential distribution with rate $1/\gamma$. This reflects that Pareto belongs to an exponential family with sufficient statistic $\log x$. The log-transform thus provides a natural bridge between power-law and exponential-family distributions; we develop the formal theory in Section 3.

Regular variation. Heavy-tailed distributions are characterized by *regular variation*: $\mathbb{P}(X > tx)/\mathbb{P}(X > t) \to x^{-\alpha}$ as $t \to \infty$, with tail index $\alpha > 0$. This captures the essential property of polynomial tail decay without requiring the exact Pareto form. For $X \sim \text{Pareto}(\gamma)$, the corresponding tail index is $\alpha = 1/\gamma$; Student-t, Burr, and log-gamma are also regularly varying. The log-transform maps any regularly varying distribution to one with exponential-type tails; see Proposition 3.8.

2.2. Generative Models

Normalizing flows. Normalizing flows (Rezende & Mohamed, 2015; Dinh et al., 2017) learn an invertible transformation $T : \mathbb{R}^d \to \mathbb{R}^d$ mapping a base distribution (typically Gaussian) to the target. The change-of-variables formula gives exact likelihoods. However, for stability, flow architectures use Lipschitz-continuous components.

Diffusion and flow matching. Denoising diffusion models (Ho et al., 2020; Song et al., 2021b) and flow matching (Lipman et al., 2023) construct a path from data to noise and learn to reverse it. Given data X_0 and noise $X_1 \sim \mathcal{N}(0, I)$, the interpolant is:

$$X_t = \alpha_t X_0 + \beta_t X_1,$$

with schedules satisfying $(\alpha_0, \beta_0) = (1, 0)$ and $(\alpha_1, \beta_1) = (0, 1)$. A neural network learns the velocity field $v_\theta(x_t, t)$ or score $\nabla \log p_t(x_t)$, enabling generation by integrating from noise to data.

2.3. The Lipschitz Barrier

Jaini et al. (2020) proved that Lipschitz maps preserve tail type:

Theorem 2.1 (Jaini et al. (2020))**.** *If Z is light-tailed and T is Lipschitz, then $T(Z)$ is light-tailed.*

Neural networks composed of linear layers and standard activations (ReLU, tanh, sigmoid) are Lipschitz. Thus, standard normalizing flows cannot map Gaussian noise to heavy-tailed outputs.

2.4. Prior Approaches

Heavy-tailed base distributions. Tail-Adaptive Flows (TAF; Jaini et al., 2020) use Student-t base distributions. Extensions include marginal TAF (Laszkiewicz et al., 2022) and generalized TAF. Heavy-tailed diffusion (Pandey et al., 2024) replaces Gaussian noise with Student-t. These methods address the Lipschitz barrier but can suffer training instability.

Tail-modifying transforms. Tail Transform Flows (TTF; Hickling & Prangle, 2025) add explicit layers that map Gaussian tails to power-law tails using erfc-based transformations. This requires estimating tail parameters per dataset via the Hill estimator.

In the next section, we present an alternative: apply the log-transform to data *before* training, work entirely in log-space, then exponentiate samples *after* generation. This parameter-free approach requires no tail estimation and we show that it circumvents the Lipschitz barrier.

3. Tail Annealing in Log-Space

This section develops the theoretical foundation for log-space flow matching. The log-transform compresses heavy tails into a light-tailed regime where standard methods apply, and the induced dynamics in original space implement *tail annealing*: power transformations that continuously adjust the tail index from heavy to light.

The principle of annealing complex structure through a sequence of easier problems is well-established in machine learning. Noise annealing in score-based models (Song & Ermon, 2019) gradually reduces noise levels to enable learning across scales. Curriculum learning (Bengio et al., 2009) trains on easy examples before hard ones. Our tail annealing follows the same principle: rather than directly modeling heavy-tailed data (hard), we work in log-space where tails are light (easy), and the power transformation $X_0^{\alpha_t}$ provides a principled path through intermediate tail weights. Section 4 presents the complete algorithm.

3.1. The Soft-Log Transform

We work with the *soft-log* transform $\phi : \mathbb{R} \to \mathbb{R}$ defined componentwise by:

$$\phi(x) = \operatorname{sign}(x) \cdot \log(1 + |x|); \qquad \phi^{-1}(y) = \operatorname{sign}(y) \cdot (e^{|y|} - 1).$$

Unlike the standard logarithm, ϕ is smooth at the origin ($\phi'(0) = 1$) and defined on all of $\mathbb{R}$. For large $|x|$, we have $\phi(x) \approx \operatorname{sign}(x) \log |x|$, so tail behavior matches the logarithm.

Relation to Box-Cox. Transforming heavy-tailed data through a log transform is a known trick in the statistical litterature. The soft-log is inspired by, but distinct from, the Box-Cox family (Box & Cox, 1964). Recall Box-Cox: $\phi_\lambda(x) = (x^\lambda - 1)/\lambda$ for $\lambda \neq 0$, and $\phi_0(x) = \log x$ for $x > 0$. Our transform differs in two ways: it extends to all of $\mathbb{R}$ via the signed construction, and uses $\log(1+|x|)$ rather than $\log |x|$ for smoothness at zero. Asymptotically, both behave as $\log |x|$ for large $|x|$. Methods like TTF (Hickling & Prangle, 2025) use power transforms with $\lambda = 1/\nu$ estimated via the Hill estimator. In our approach, numerical computation make no use of numerical values of the tail estimate, which are notoriously unstable: the same transform applies regardless of the true tail index.

The soft-log maps heavy tails to light tails. If $X \sim \text{Pareto}(\gamma)$ with $\mathbb{P}(X > t) = t^{-1/\gamma}$ (so $\gamma > 0$ is the shape parameter and $1/\gamma$ is the tail index), then for large y:

$$\mathbb{P}(\phi(X) > y) = \mathbb{P}(X > e^y - 1) \approx e^{-y/\gamma},$$

i.e., $\tilde{X} = \phi(X)$ has exponential-type tails. More generally, for any distribution in the Fréchet domain of attraction, ϕ yields a distribution with exponential or lighter tails.

Since ϕ is a diffeomorphism, densities transform via the standard change of variables:

$$p_{\tilde{X}}(\tilde{x}) = p_X(\phi^{-1}(\tilde{x})) \cdot |(\phi^{-1})'(\tilde{x})| = p_X(\phi^{-1}(\tilde{x})) \cdot e^{|\tilde{x}|}.$$

3.2. Induced Process in Original Space

We apply flow matching to the transformed data $\tilde{X}_0 = \phi(X_0)$, interpolating with Gaussian noise $\tilde{X}_1 \sim \mathcal{N}(0, I_d)$ via $\tilde{X}_t = \alpha_t \tilde{X}_0 + \beta_t \tilde{X}_1$ (see Section 4 for the complete framework). Here we analyze the induced process in original space.

Applying ϕ^{-1} to the interpolant yields, for large positive values where $\phi \approx \log$:

$$X_t := \phi^{-1}(\tilde{X}_t) \approx X_0^{\alpha_t} \cdot e^{\beta_t \tilde{X}_1}. \tag{1}$$

This factorizes into a power-transformed data term $X_0^{\alpha_t}$ and a log-normal noise term $e^{\beta_t \tilde{X}_1}$. Although $e^{\beta_t \tilde{X}_1}$ is itself heavy-tailed in the MGF sense, it has extreme-value index $\gamma = 0$ (Gumbel MDA): $\mathbb{P}(Y_t > t) \sim e^{-c(\log t)^2}$ decays faster than any power law and all moments are finite. By Breiman's lemma it does not contribute a polynomial tail of its own to the product:

Proposition 3.1 (Product Preserves Regular Variation; Breiman)**.** *Let X be regularly varying with index $-\alpha$ ($\alpha > 0$), and let $Y > 0$ be independent with $\mathbb{E}[Y^p] < \infty$ for all $p > 0$ (e.g., any variable in the Gumbel MDA, such as log-normal). Then XY is regularly varying with index $-\alpha$.*

This classical result (see Resnick (1987)) ensures that the induced process X_t inherits its polynomial-tail behavior from $X_0^{\alpha_t}$, not from $e^{\beta_t \tilde{X}_1}$ (whose extreme-value index is 0). The power transformation is therefore what drives tail annealing.

3.3. Score Functions in Log-Space

The score function $\nabla_{\tilde{x}} \log p_t(\tilde{x})$ in log-space has fundamentally different behavior than in original space.

Proposition 3.2 (Log-Space Score of Pareto)**.** *Let $X \sim$ Pareto(γ) (so $\bar{F}_X(t) = t^{-1/\gamma}$). Then $\tilde{X} = \phi(X)$ has approximately exponential tails: for large $\tilde{x}$,*

$$\nabla_{\tilde{x}} \log p_{\tilde{X}}(\tilde{x}) \approx -\frac{1}{\gamma}.$$

The score is approximately constant in the tails.

Proof. For large x, $\phi(x) \approx \log x$, so $\tilde{X} \approx \log X \sim \text{Exp}(1/\gamma)$. The exponential density is $p(\tilde{x}) \propto e^{-\tilde{x}/\gamma}$, giving $\nabla_{\tilde{x}} \log p(\tilde{x}) = -1/\gamma$. □

Pareto score. By the *Pareto score* we mean the score of the Pareto density $p_X(x) = (1/\gamma)\, x^{-1/\gamma-1}$ in original space:

$$s_X(x) = \nabla_x \log p_X(x) = -\frac{1/\gamma + 1}{x},$$

which diverges as $x \to 0^+$ and decays only as $1/x$ in the tails. The log-space score in Proposition 3.2 is by contrast bounded by $1/\gamma$ everywhere, so the velocity field that flow matching has to regress is Lipschitz in $\tilde{x}$. This score boundedness is what enables the standard MLP velocity network to learn a target with regularly varying tails in the original space.

Proposition 3.3 (Score of Noised Log-Data)**.** *For the interpolant $\tilde{X}_t = \alpha_t \tilde{X}_0 + \beta_t \tilde{X}_1$ with $\tilde{X}_0 = \phi(X_0)$ having exponential-type tails and $\tilde{X}_1 \sim \mathcal{N}(0, I_d)$, the marginal score satisfies:*

$$\nabla_{\tilde{x}_t} \log p_t(\tilde{x}_t) = -\frac{\hat{\tilde{x}}_1(\tilde{x}_t, t)}{\beta_t},$$

where $\hat{\tilde{x}}_1(\tilde{x}_t, t) := \mathbb{E}[\tilde{X}_1 \mid \tilde{X}_t = \tilde{x}_t]$ is the conditional expectation of the noise.

This is the standard score-denoiser relationship. The key difference from original-space diffusion is that the transformed data distribution $p_{\tilde{X}_0}$ is light-tailed, so:

1. The score $\nabla \log p_t$ is well-behaved for all t
2. The denoiser $\hat{\tilde{x}}_1(\tilde{x}_t, t)$ has bounded outputs
3. Standard neural network architectures suffice

3.4. Tail Behavior Under Power Transformation

The induced process (1) involves the power-transformed data $X_0^{\alpha_t}$. We characterize its distribution. Note that this analysis describes the *decoded* interpretation: in practice, all computation occurs in log-space where distributions are light-tailed. The heavy-tailed structure below is what we would observe if we applied ϕ^{-1} to the log-space interpolant at each t.

Lemma 3.4 (Power Transformation Preserves Pareto Family)**.** *Let $X_0 \sim$ Pareto(γ) with density $p_{X_0}(x) = (1/\gamma)\, x^{-1/\gamma-1}$ for $x \geq 1$, so $\gamma > 0$ is the shape parameter (larger γ means heavier tails). For any $\alpha \in (0, \infty)$, $X_0^\alpha \sim$ Pareto$(\gamma\alpha)$: the shape parameter scales linearly with the exponent.*

Proof. The survival function of $Y = X_0^\alpha$ is $\mathbb{P}(Y > y) = \mathbb{P}(X_0 > y^{1/\alpha}) = y^{-1/(\gamma\alpha)}$ for $y \geq 1$, which is the survival function of Pareto$(\gamma\alpha)$. $\square$

Remark 3.5 (Parametrization). We use γ for the GPD shape parameter / extreme-value index, so $\mathbb{P}(X > t) = t^{-1/\gamma}$. The corresponding tail index (Hill estimator output, the standard EVT parameter) is $1/\gamma$. Under Lemma 3.4 the power transform X_0^α has shape parameter $\gamma\alpha$ and tail index $1/(\gamma\alpha)$.

Corollary 3.6 (Tail Lightening Along Forward Process)**.** *Along the forward process with α_t decreasing from 1 to 0:*

- *At $t = 0$: $\alpha_0 = 1$; $X_0^{\alpha_0} = X_0 \sim$ Pareto(γ) (original heavy tails)*
- *At intermediate t: $X_0^{\alpha_t} \sim$ Pareto$(\gamma\alpha_t)$ (lighter tails as $\alpha_t \to 0$)*
- *As $\alpha_t \to 0$: $X_0^{\alpha_t} \to 1$ almost surely (degenerate)*

The full process $X_t = X_0^{\alpha_t} \cdot e^{\beta_t \tilde{X}_1}$ transitions from regularly-varying Pareto (shape γ, tail index $1/\gamma$) to log-normal (extreme-value index 0, no polynomial tail).

This is what we frame as the tail annealing mechanism: the power transformation $X_0^{\alpha_t}$ preserves the Pareto family while continuously adjusting the shape parameter from γ (heavy) toward 0 (light) as $\alpha_t \to 0$. We state Lemma 3.4 for $\alpha \in (0, \infty)$ rather than the forward-process range $\alpha_t \in (0, 1]$, because the same identity appears in Proposition 3.9 for the regularly varying extension.

3.5. Extension to Regularly Varying Distributions

The Pareto-exponential correspondence extends to the broader class of regularly varying distributions.

Definition 3.7 (Regularly Varying)**.** A cumulative distribution function F on $\mathbb{R}$ is regularly varying with index $-\alpha$ (written $F \in RV_{-\alpha}$) if its survival function $\bar{F}(x) = 1 - F(x)$ satisfies:

$$\lim_{t \to \infty} \frac{\bar{F}(tx)}{\bar{F}(t)} = x^{-\alpha}$$

for all $x > 0$. Here $\alpha > 0$ is the (polynomial) tail index, in the standard EVT convention; for $X \sim$ Pareto(γ), $\alpha = 1/\gamma$.

Regular variation captures the essential property of polynomial tail decay without requiring the exact Pareto form:

Distribution	Tail behavior	Index
Pareto(γ)	$\bar{F}(x) = x^{-1/\gamma}$	$-1/\gamma$
Student-$t(\nu)$	$\bar{F}(x) \sim c \cdot x^{-\nu}$	$-\nu$
Burr(c, k)	$\bar{F}(x) \sim c' \cdot x^{-ck}$	$-ck$
Log-gamma	$\bar{F}(x) \sim c'' \cdot x^{-\alpha}(\log x)^\beta$	$-\alpha$

Proposition 3.8 (Log-Transform of Regularly Varying; informal)**.** *If X is regularly varying with index $-\alpha$, then $\tilde{X} = \phi(X)$ has exponential-type right tail with rate α:*

$$-\log \mathbb{P}(\tilde{X} > z) \;=\; \alpha\, z + o(z) \quad \textit{as } z \to \infty.$$

A precise two-sided Potter-bound version is stated and proved in Appendix C (Proposition C.1). The intuition is direct: writing the regularly varying tail as $\bar{F}_X(x) =$

$x^{-\alpha}L(x)$ with L slowly varying, and using $\phi^{-1}(z) = e^z - 1 \sim e^z$, gives $\mathbb{P}(\tilde{X} > z) = e^{-\alpha z}L(e^z)$, and the slowly varying factor only contributes a subexponential correction $\log L(e^z) = o(z)$.

This proposition is the conceptual reason flow matching in log-space is well-posed for the full class of regularly varying distributions, not just exact Pareto. Up to slowly-varying corrections, $\tilde{X}$ has the same exponential rate $e^{-\alpha z}$ as $\log X$, which equals $e^{-z/\gamma}$ when $X \sim \text{Pareto}(\gamma)$. The score in log-space is therefore bounded in the tails and standard diffusion methods apply.

Beyond regular variation. The transform is also beneficial for subexponential distributions outside the Fréchet domain. For Weibull tails $\bar{F}(x) \sim \exp(-x^\beta)$ with $\beta < 1$, the log-transform yields a doubly-exponential tail $\mathbb{P}(\tilde{X} > z) \sim \exp(-(e^z - 1)^\beta)$, which is well within the regime where standard flow matching has no difficulty. For log-normal data, $\phi(X) \approx \log X$ is exactly Gaussian, so the log-space target is the ideal case for FM. Light-tailed margins are addressed by the adaptive variant of Section 4.2.

Proposition 3.9 (Power Transformation of Regularly Varying)**.** *If $X \in RV_{-\alpha}$, then X^β for $\beta \in (0, \infty)$ is regularly varying with index $-\alpha/\beta$.*

Proof. For $Y = X^\beta$, we have $\mathbb{P}(Y > ty)/\mathbb{P}(Y > t) = \mathbb{P}(X > (ty)^{1/\beta})/\mathbb{P}(X > t^{1/\beta}) \to y^{-\alpha/\beta}$ as $t \to \infty$, by regular variation of X. This is the regularly varying analogue of Lemma 3.4: power transformations remap the tail index, so X^{α_t} along the forward process continuously interpolates between the original tail ($\alpha_t = 1$) and arbitrarily light tails ($\alpha_t \in [0, 1)$]). □

This confirms that the tail-annealing mechanism of Lemma 3.4 extends beyond exact Pareto to the full class of regularly varying distributions.

3.6. Multivariate Extension

In the multivariate setting, $X \in \mathbb{R}^d$, we apply ϕ coordinate-wise:

$$\phi(\mathbf{x}) = (\phi(x_1), \ldots, \phi(x_d)), \qquad \tilde{X} = \phi(X).$$

Each coordinate of $\tilde{X}$ inherits the exponential-rate tail of Proposition 3.8, so the marginals of $\tilde{X}$ all live in the Gumbel domain. Because ϕ is a diffeomorphism with diagonal Jacobian $J_\phi(\mathbf{x}) = \text{diag}((1 + |x_i|)^{-1})$, the dependence structure (copula) of X is preserved exactly by $\tilde{X}$: the log-transform is a marginal operation. Interpolating $\tilde{X}_t = \alpha_t \tilde{X}_0 + \beta_t \tilde{X}_1$ with $\tilde{X}_1 \sim \mathcal{N}(0, I_d)$ is therefore well-posed even when X has strong extremal dependence: the velocity network sees only the light-tailed transformed variates and never has to extrapolate to the polynomial regime.

Heterogeneous margins. When the coordinates of X have different tail behaviour (heavy-tailed in some, light-tailed in others), applying ϕ to every coordinate slightly distorts the light-tailed margins. The adaptive variant of Section 4.2 addresses this by Hill-gating the transform coordinate-wise; a continuous generalization of the soft-log via a scale parameter s_2, of which the adaptive variant is the binary $s_2 \in \{0, 1\}$ instance, is developed in Appendix D.

3.7. Circumventing the Lipschitz Barrier

The Lipschitz barrier of Jaini et al. (2020) states that Lipschitz maps preserve tail type: a normalizing flow with Lipschitz layers cannot map Gaussian noise to heavy-tailed outputs. Our construction circumvents this by placing the non-Lipschitz transformation (ϕ^{-1}) *outside* the learned dynamics. The flow operates entirely in log-space where both endpoints (exponential-type transformed data and Gaussian noise) have light tails. Heavy tails emerge only upon applying ϕ^{-1} at the final sampling step.

4. Algorithm

We present the complete algorithm for log-space generative modeling. The method requires the three following modifications to standard flow matching: (1) transform data via ϕ before training (with a Hill-based diagnostic deciding which coordinates to transform), (2) train a velocity network on the transformed variables, and (3) apply ϕ^{-1} to the corresponding coordinates of generated samples. The Hill diagnostic distinguishes *Log-FM* from a standard coordinate-wise log-transform: it lets the method handle heterogeneous-margin data with a single Hill estimate per coordinate at fit time.

4.1. Flow Matching Framework

We use the flow matching framework (Lipman et al., 2023), which can also be viewed through the lens of stochastic interpolants (Albergo & Vanden-Eijnden, 2023).

Forward process. Given log-transformed data $\tilde{X}_0 = \phi(X_0)$ and noise $\tilde{X}_1 \sim \mathcal{N}(0, I_d)$, define the interpolant:

$$\tilde{X}_t = \alpha_t \tilde{X}_0 + \beta_t \tilde{X}_1,$$

where (α_t, β_t) are differentiable schedules satisfying boundary conditions $(\alpha_0, \beta_0) = (1, 0)$ and $(\alpha_1, \beta_1) = (0, 1)$. At $t = 0$, $\tilde{X}_0$ is the transformed data; at $t = 1$, $\tilde{X}_1$ is pure Gaussian noise. The conditional distribution is Gaussian: $q_{t|0}(\tilde{x}_t \mid \tilde{x}_0) = \mathcal{N}(\tilde{x}_t; \alpha_t \tilde{x}_0, \beta_t^2 I_d)$.

Velocity field. Differentiating the interpolant gives the per-trajectory time derivative

$$\dot{\tilde{X}}_t \;=\; \dot{\alpha}_t \tilde{X}_0 + \dot{\beta}_t \tilde{X}_1,$$

where $\dot{\alpha}_t = d\alpha_t/dt$ and $\dot{\beta}_t = d\beta_t/dt$. The corresponding *marginal* velocity field, $v_t(\tilde{x}_t) := \mathbb{E}[\dot{\alpha}_t \tilde{X}_0 + \dot{\beta}_t \tilde{X}_1 \mid \tilde{X}_t = \tilde{x}_t]$, is what a neural network v_θ regresses against the per-trajectory target:

$$\mathcal{L}(\theta) = \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\tilde{X}_0,\tilde{X}_1} \left[\left\| v_\theta(\tilde{X}_t, t) - (\dot{\alpha}_t \tilde{X}_0 + \dot{\beta}_t \tilde{X}_1)\right\|^2\right].$$

Connection to score and denoiser. The velocity field relates to the score $\nabla \log p_t$ and denoiser $\hat{\tilde{x}}_0(\tilde{x}_t, t) := \mathbb{E}[\tilde{X}_0 \mid \tilde{X}_t = \tilde{x}_t]$ via:

$$v_t(\tilde{x}_t) = \dot{\alpha}_t \hat{\tilde{x}}_0(\tilde{x}_t, t) + \dot{\beta}_t \hat{\tilde{x}}_1(\tilde{x}_t, t)$$
$$\nabla_{\tilde{x}_t} \log p_t(\tilde{x}_t) = -\frac{\hat{\tilde{x}}_1(\tilde{x}_t, t)}{\beta_t},$$

where $\hat{\tilde{x}}_1 = (\tilde{x}_t - \alpha_t \hat{\tilde{x}}_0)/\beta_t$. See Appendix B for derivations.

Schedule choices. The interpolation schedule (α_t, β_t) controls the path from data to noise. Common choices include the *linear* schedule (optimal transport paths), *variance-preserving* (VP) schedules, and *quadratic* schedules that anneal tails more aggressively; see Table 6 in Appendix B. We use the linear schedule throughout; preliminary tests showed minimal sensitivity to schedule choice, consistent with prior flow matching work (Lipman et al., 2023).

Connection to tail annealing. Recall from Section 3 that the induced process in original space is approximately $X_t \approx X_0^{\alpha_t} \cdot e^{\beta_t \tilde{X}_1}$. The schedule α_t thus controls the rate of tail annealing: faster decay of α_t (e.g., quadratic) anneals tails more aggressively early in the process, while slower decay (e.g., VP polynomial) preserves heavier tails longer.

4.2. Adaptive Coordinate Selection

Applying the soft-log to a light-tailed coordinate compresses its bulk for no benefit. We therefore select *per coordinate* whether to transform, using a Hill estimate on the training marginals. Let $\hat{\alpha}_j$ be the Hill estimator on the upper order statistics of $\{x_i^{(j)}\}_{i=1}^N$ (an estimate of the polynomial tail index, with $\hat{\alpha}_j = 1/\hat{\gamma}_j$ under the Pareto-shape parametrization). Define the mask

$$m_j \;=\; \mathbf{1}\{\hat{\alpha}_j \le \alpha_{\max}\}, \qquad \alpha_{\max} = 4, \tag{2}$$

and the per-coordinate transform $\Phi : \mathbb{R}^d \to \mathbb{R}^d$ by

$$\Phi(\mathbf{x})_j = \begin{cases} \phi(x_j) & \text{if } m_j = 1, \\ x_j & \text{otherwise,} \end{cases} \tag{3}$$

Algorithm 1 Log-FM: Training

Input: Dataset $\{\mathbf{x}_i\}_{i=1}^N \subset \mathbb{R}^d$, velocity network v_θ, schedule (α_t, β_t), iterations T, Hill threshold $\alpha_{\max} = 4$
Fit transform: for each coordinate j, compute Hill estimate $\hat{\alpha}_j$ on $\{x_i^{(j)}\}_i$; set mask m_j via (2)
Preprocess: $\tilde{\mathbf{x}}_i \leftarrow \Phi(\mathbf{x}_i)$ for all i {coordinate-wise; identity where $m_j = 0$}
for iter $= 1$ **to** T **do**
  Sample batch $\{\tilde{\mathbf{x}}_0^{(j)}\}$ from $\{\tilde{\mathbf{x}}_i\}$
  Sample $\tilde{\mathbf{x}}_1^{(j)} \sim \mathcal{N}(0, I_d)$
  Sample $t \sim \text{Uniform}[0,1]$
  Form interpolant $\tilde{\mathbf{x}}_t^{(j)} \leftarrow \alpha_t \tilde{\mathbf{x}}_0^{(j)} + \beta_t \tilde{\mathbf{x}}_1^{(j)}$
  Compute target $u^{(j)} \leftarrow \dot{\alpha}_t \tilde{\mathbf{x}}_0^{(j)} + \dot{\beta}_t \tilde{\mathbf{x}}_1^{(j)}$
  Loss $\mathcal{L} \leftarrow \frac{1}{|\text{batch}|}\sum_j \|v_\theta(\tilde{\mathbf{x}}_t^{(j)}, t) - u^{(j)}\|^2$
  Update θ via gradient descent on $\mathcal{L}$
end for
Output: Trained v_θ, mask $(m_j)_{j=1}^d$

with inverse $\Phi^{-1}(\tilde{\mathbf{x}})_j = \phi^{-1}(\tilde{x}_j)$ when $m_j = 1$ and $\tilde{x}_j$ otherwise. The threshold $\alpha_{\max} = 4$ is not sensitive: on heterogeneous-margin data with Pareto coordinates ($\hat{\alpha} \approx$ 1.5–2.5) and Gaussian coordinates ($\hat{\alpha} \approx 6$), any threshold in $[3, 5]$ yields the same mask. Setting $m_j \equiv 1$ recovers the uniform-ϕ variant analyzed in Section 3; the adaptive rule is the discrete instance of a continuous parametrized soft-log family φ_{s_2} with $s_2^{(j)} \in \{0, 1\}$ chosen by the Hill diagnostic, developed in Appendix D. We use Φ (with the Hill-based mask) as our default; we also report the uniform variant as an ablation in Section 5.3.

4.3. Training

Algorithm 1 summarizes the training procedure. The only departure from standard flow matching is the initial log-transform of data (with the Hill diagnostic of Section 4.2).

4.4. Sampling

Generation reverses the flow: integrate the learned velocity field from noise ($t = 1$) to data ($t = 0$), then apply the inverse transform. The velocity field satisfies the ODE:

$$\frac{d\tilde{X}_t}{dt} = v_\theta(\tilde{X}_t, t),$$

which we solve backward in time. Algorithm 2 uses Euler discretization; higher-order solvers (Heun, RK4) improve sample quality with fewer steps.

4.5. Practical Considerations

Output clamping (optional). The inverse transform $\phi^{-1}(\tilde{x}) = \text{sign}(\tilde{x})(e^{|\tilde{x}|} - 1)$ is exponential, so small errors

Algorithm 2 Log-FM: Sampling

Input: Trained v_θ, mask (m_j), steps K, optional clamp c (default $c = \infty$)
Sample $\tilde{\mathbf{x}} \sim \mathcal{N}(0, I_d)$ {initialize at $t = 1$}
$\Delta t \leftarrow 1/K$
for $k = K$ **downto** 1 **do**
  $t \leftarrow k/K$
  $\tilde{\mathbf{x}} \leftarrow \tilde{\mathbf{x}} - \Delta t \cdot v_\theta(\tilde{\mathbf{x}}, t)$ {Euler step}
end for
(optional) clamp coordinates with $m_j = 1$: $\tilde{x}_j \leftarrow \mathrm{clamp}(\tilde{x}_j, -c, c)$ {only needed for $\alpha < 1$}
Output: $\mathbf{x} = \Phi^{-1}(\tilde{\mathbf{x}})$ {coordinate-wise inverse from (3)}

deep in the tail of $\tilde{x}$ are exponentially amplified in x. An optional clamp $\tilde{x}_j \leftarrow \mathrm{clamp}(\tilde{x}_j, -c, c)$ before ϕ^{-1} provides a numerical safeguard in extremely pathological regimes only (typically $\alpha < 1$: Cauchy and heavier, e.g. Hickling's $\nu = 1/2$ Student-t baseline). For every configuration in our main benchmark ($\alpha \geq 1.5$) the clamp is inert at $c \geq 10$, and we report all main-benchmark numbers without clamping; see the ablation in Section 5.3.

Number of integration steps. We use $K = 100$ Euler steps for all experiments. Higher-order solvers (Heun, RK4) can reduce step count but Euler suffices.

Architecture. We use MLPs with sinusoidal time embeddings (each scalar t is mapped to $[\sin(2\pi\omega_k t), \cos(2\pi\omega_k t)]_{k=1}^{K_\omega}$ with geometric frequencies, exactly as in Ho et al. (2020)): 4 hidden layers of width 256 with SiLU activations. No architectural modifications are needed for heavy tails; the log-transform handles tail behavior.

Likelihood evaluation. We report NLL using the continuous change-of-variables formula for continuous normalizing flows. The trace of the velocity Jacobian is estimated stochastically via Hutchinson's trick with $K = 10$ Rademacher projections, and the augmented ODE is integrated by the adaptive Dormand–Prince `dopri5` scheme with $\mathrm{atol} = \mathrm{rtol} = 10^{-5}$. The transform itself contributes the closed-form Jacobian term $\sum_{j: m_j = 1} \log(1 + |x_j|)$, which is exact and negligible to compute.

Training. AdamW optimizer with learning rate 5×10^{-3} and weight decay 10^{-5}. We use full-batch gradient descent with gradient clipping at 10.0 (before inverse transform). Training runs for up to 5000 epochs with early stopping (patience 100) based on validation loss.

4.6. Extensions

Signed and multivariate data. The soft-log $\phi(x) = \mathrm{sign}(x) \log(1 + |x|)$ applies directly to signed data and acts coordinate-wise on $\mathbb{R}^d$ via Φ; the dependence structure is preserved exactly on transformed coordinates and learned implicitly by the velocity field.

Arcsinh variant. Replacing ϕ by $\mathrm{arcsinh}(x) = \log(x + \sqrt{1 + x^2})$ gives the Arcsinh-FM variant evaluated in Section 5. Both transforms have the same logarithmic asymptotic, so Propositions 3.8–3.9 apply unchanged. They differ only in regularity at the origin: arcsinh is C^∞ on $\mathbb{R}$, while ϕ is C^1 but not C^2 at 0 (its second derivative jumps from $+1$ to -1 at the origin), which isolates whether higher-order regularity matters in practice.

5. Experiments

We evaluate Log-FM against state-of-the-art heavy-tailed generative models on a controlled multivariate benchmark with known marginal tails and known copula structure. Our goals are to test whether the log-transform (i) preserves the dependence structure of the data, (ii) yields stable training across heavy-tail regimes, (iii) recovers risk-relevant tail metrics (VaR_{99}, CVaR_{99}, $Q_{99.9}$) that are sensitive to the deep tail. A Student-t benchmark from (Hickling & Prangle, 2025) is detailed for continuity in Appendix E.1; real-data results on Fama–French are deferred to Appendix E.2.

5.1. Setup

Data. We use 144 configurations covering three copula families, three dependence strengths, four dimensions, and four tail indices:

- **Copulas:** Gaussian copula (asymptotic independence), Gumbel copula (symmetric upper tail dependence) at Kendall's $\tau \in \{0.25, 0.5, 0.75\}$, and Hüsler–Reiss copula with AR(1) variogram $\Gamma_{ij} = 2(1 - \rho^{|i-j|})$ at $\rho \in \{0.1, 0.5, 0.9\}$.
- **Margins:** 70% symmetrized Pareto with tail index $\alpha \in \{1.5, 1.75, 2.0, 2.5\}$ (smaller = heavier), 30% standard Gaussian.
- **Dimensions:** $d \in \{10, 20, 50, 100\}$.
- **Samples:** $n_{\text{train}} = 10{,}000$, $n_{\text{val}} = 5{,}000$, $n_{\text{test}} = 20{,}000$.

For each configuration we train 5 methods $\times$ 20 independent replications, giving 14,400 trained models in total.

Methods. Log-FM (default, with Hill-gated transform of Section 4.2, $\alpha_{\max} = 4$); Log-FM (uniform, $m_j \equiv 1$);

Arcsinh-FM (uniform, with $\phi(x) = \text{arcsinh}(x)$); TTF and TTFfix (Hickling & Prangle, 2025); gTAF (Jaini et al., 2020). All velocity / coupling networks share the same MLP backbone (4 layers, width 256, SiLU); hyperparameters per dimension were selected by Optuna on a representative configuration (Gumbel, $\tau = 0.5$, $\alpha = 2.0$) and reused across the grid for that dimension.

Metrics. We split metrics by margin type and by joint structure:

- *Marginal:* W_1^P (mean W_1 over Pareto coordinates), W_1^G (over Gaussian coordinates), Hill estimator on Pareto margins, VaR_{99} / CVaR_{99} relative errors, extreme-quantile errors $Q_{99.5}$, $Q_{99.9}$.
- *Joint:* Absolute Kendall Error (AKE), angular W_2 (sliced Wasserstein on the empirical angular measure of the top-$\sqrt{n}$ extremes), sliced Wasserstein on the full data, and energy distance.

We report medians over the 20 replications; "div." marks runs where the model diverged ($W_1 > 10^3$).

5.2. Main Results

Table 1 reports the headline tail metric, W_1^P, averaged over the two regular copulas (Gumbel + Gaussian) at the four tail indices and four dimensions. Log-FM is best in 13 out of 16 cells; Arcsinh-FM picks up the remaining 3 at $d = 50$. Both FM variants are substantially more stable than the baselines, especially for $\alpha = 1.5$.

Risk metrics and dependence. Table 2 summarizes the global picture, pooling over both copulas, the three dependence strengths, and all four tail indices. Log-FM is best on every tail-quality and risk metric (W_1^P, CVaR_{99}, $Q_{99.9}$, sliced and energy distances). On the multivariate-dependence metrics, TTF/TTFfix retain a small edge at $d = 50, 100$ while Log-FM is competitive at $d = 10, 20$. On W_1^G (Gaussian margins), the Hill-gated default of Log-FM closes most of the gap with TTF (further breakdown in Section 5.3).

Stability. A practitioner cares not only about median performance but about how often the model fails catastrophically. Table 3 reports, by tail index, the fraction of runs with $W_1^P > 1$ on Gumbel+Gaussian. Log-FM and Arcsinh-FM never exceed 14% even in the most adversarial setting ($\alpha = 1.5$, $d = 10$); both reach 0% for $\alpha \geq 1.75$. gTAF and TTF suffer divergence rates above 30% in higher dimensions.

Table 1. Median W_1^P across Gumbel + Gaussian copulas (lower is better, bold = best). Log-FM dominates at every tail index and dimension; the gap widens for heavier tails ($\alpha = 1.5$).

Method	d=10	d=20	d=50	d=100
$\alpha = 1.5$				
Log-FM	**0.522**	**0.451**	0.534	**0.525**
Arcsinh-FM	0.561	0.550	**0.534**	0.592
TTFfix	1.149	0.963	0.867	1.046
TTF	0.758	0.938	0.758	0.919
gTAF	0.534	0.583	0.761	3.320
$\alpha = 1.75$				
Log-FM	**0.254**	**0.226**	**0.269**	**0.300**
Arcsinh-FM	0.262	0.284	0.289	0.316
TTFfix	0.442	0.417	0.391	0.481
TTF	0.318	0.373	0.354	0.401
gTAF	0.270	0.301	0.335	0.934
$\alpha = 2.0$				
Log-FM	**0.153**	**0.146**	0.195	**0.196**
Arcsinh-FM	0.166	0.162	**0.174**	0.212
TTFfix	0.235	0.222	0.220	0.294
TTF	0.182	0.203	0.207	0.252
gTAF	0.171	0.174	0.198	0.219
$\alpha = 2.5$				
Log-FM	**0.080**	**0.074**	**0.090**	**0.108**
Arcsinh-FM	0.082	0.080	0.100	0.116
TTFfix	0.108	0.107	0.117	0.174
TTF	0.083	0.106	0.113	0.131
gTAF	0.088	0.086	0.106	0.225

5.3. Ablations

We isolate three design choices: the Hill gating (adaptive vs uniform), the sampling clamp c, and the ODE solver step count.

Hill-gated default vs uniform. Table 4 compares the default Log-FM (with Hill gating, Section 4.2) against the uniform variant and the TTFfix baseline on Gumbel data ($d = 20$, $\tau = 0.5$, 20 reps). The Hill diagnostic always selects exactly the heavy-tailed coordinates (14 Pareto vs 6 Gaussian out of 20): the threshold $\alpha_{\max} = 4$ is non-sensitive given $\hat{\alpha} \approx 1.5$–2.5 for Pareto and ≈ 6 for Gaussian. The adaptive variant improves W_1^G by 30–45% with no W_1^P cost.

Clamping (optional). The sampling clamp c is an *optional* numerical safeguard applied in log-space (corresponding to $|x| \lesssim e^c$ in data space); it is not used by default. Table 5 reports the relative change in W_1^P across c values. For every configuration in our main benchmark ($\alpha \geq 1.5$) the clamp is inert at $c \geq 10$: metrics are bitwise identical to the no-clamp case. The clamp only becomes relevant in extremely pathological tail regimes ($\alpha < 1$, e.g. in $\nu = 1/2$ Student-t in Appendix E.1), where the population W_1 is undefined and exponential error amplification through ϕ^{-1}

Table 2. Median across all Gumbel+Gaussian configurations (480 values per cell). Bold = best per metric per dimension.

Method	d=10	d=20	d=50	d=100
W_1^P **(Pareto margins)**				
Log-FM	**0.207**	**0.187**	**0.221**	**0.233**
Arcsinh-FM	0.218	0.216	0.232	0.264
TTFfix	0.322	0.322	0.308	0.395
TTF	0.250	0.304	0.295	0.349
gTAF	0.262	0.304	0.540	0.704
CVaR$_{99}$ (Pareto margins)				
Log-FM	**0.228**	**0.200**	0.257	**0.270**
Arcsinh-FM	0.246	0.224	**0.252**	0.373
TTFfix	0.880	0.689	0.644	0.817
TTF	0.494	0.729	0.687	0.680
gTAF	0.415	0.457	0.519	0.705
W_1^G **(Gaussian margins)**				
Log-FM	0.051	0.066	0.080	0.080
Arcsinh-FM	0.045	0.057	0.074	0.085
TTFfix	**0.033**	0.048	0.043	0.051
TTF	0.035	**0.035**	**0.030**	**0.047**
gTAF	0.036	0.051	0.106	div.
Angular W_2 (tail dependence)				
Log-FM	0.098	**0.066**	0.071	0.049
Arcsinh-FM	**0.098**	0.068	0.090	0.049
TTFfix	0.148	0.095	0.058	0.040
TTF	0.121	0.105	**0.046**	**0.040**
gTAF	0.167	0.104	0.061	0.098

Table 3. Fraction of runs with $W_1^P > 1$ (catastrophic failures), Gumbel + Gaussian, 20 reps. Lower is better.

Method	d=10	d=20	d=50	d=100
$\alpha = 1.5$				
Log-FM	14%	2%	4%	2%
Arcsinh-FM	11%	6%	2%	2%
TTFfix	58%	50%	36%	54%
TTF	32%	45%	36%	46%
gTAF	8%	24%	42%	59%
$\alpha = 2.0$				
Log-FM	0%	0%	0%	0%
Arcsinh-FM	0%	0%	0%	0%
TTFfix	2%	2%	1%	4%
TTF	1%	4%	12%	12%
gTAF	17%	23%	32%	33%

can produce numerical overflow rather than a meaningful sample. All headline numbers (Tables 1–3) are reported *without* clamping.

ODE solver steps. Sweeping the Euler step count $K \in \{10, 20, 50, 100, 200, 500\}$ on Gumbel ($d = 20$, $\tau = 0.5$, 10 reps) yields less than 5% variation in W_1^P; our default $K = 100$ is well within the converged regime. The full table is reported in Appendix E.3.

5.4. Discussion

The benchmark confirms the theoretical analysis, in that Log-FM has better numerical results whenever marginals are genuinely heavy-tailed ($\alpha \lesssim 3$), with "few severe divergences in all scenarios" (Table 3) as the strongest practical

Table 4. Log-FM (uniform ϕ) vs Log-FM (Hill-adaptive) vs TTFfix baseline. Gumbel, $d = 20$, $\tau = 0.5$, 20 reps, median. Bold = best per α.

α	Method	W_1^P	W_1^G	AKE
1.5	TTFfix	0.856	0.049	0.071
	Log-FM	**0.449**	0.074	**0.039**
	Log-FM (adaptive)	0.468	**0.043**	0.044
2.0	TTFfix	0.225	0.034	0.066
	Log-FM	0.128	0.040	0.031
	Log-FM (adaptive)	**0.124**	**0.024**	**0.030**
2.5	TTFfix	0.113	0.079	0.060
	Log-FM	**0.068**	0.110	**0.030**
	Log-FM (adaptive)	0.084	**0.068**	0.037

Table 5. Clamp ablation: relative W_1^P change vs. no clamping ($c = \infty$, default). Gumbel, $d = 20$, $\tau = 0.5$, 5 reps. The clamp is inert for $\alpha \geq 1.5$; it is only relevant in pathological regimes ($\alpha < 1$).

c	α=1.5	α=1.75	α=2.0	α=2.5
5	−2.0%	−0.7%	−0.6%	0.0%
10, 15, 20	0.0%	0.0%	0.0%	0.0%
∞	0.0%	0.0%	0.0%	0.0%

claim. Dependence is largely preserved by Φ's diagonal Jacobian; baselines retain only a slight angular-W_2 edge at $d \geq 50$. Real-data validation on Fama–French ($\hat{\alpha} \approx 3.7$) is in Appendix E.2.

6. Conclusion

A coordinate-wise soft-log $\phi(x) = \operatorname{sign}(x)\log(1 + |x|)$, alongside a Hill diagnostic, makes standard flow matching work for heavy-tailed data. The mechanism is tail annealing: log-transforming Pareto yields approximately exponential tails, and the induced dynamics implement power transformations $X_0^{\alpha_t}$ that continuously lighten the tail index. Across 2,880 runs (3 copulas, 4 dimensions, 4 tail indices), Log-FM dominates specialized baselines on W_1, CVaR$_{99}$, and extreme-quantile metrics, never diverges severely, and remains competitive on multivariate dependence; on Fama-French it matches the best baseline. For $\alpha < 1$ all methods struggle (W_1 is undefined); at $d \gtrsim 200$ the bottleneck shifts to architecture. Three theoretical directions stand out: a continuous variant via φ_{s_2} (Appendix D); combining the transform with discrete normalizing flows, where $|\det J_\Phi| = \prod_j (1 + |x_j|)^{-m_j}$ gives exact log-likelihood at $O(d)$ cost; and finite-sample guarantees for log-space score estimation, where bounded scores (Proposition 3.2) place the problem in the regime of existing convergence results.

Impact Statement

This work targets heavy-tailed generative modeling, with applications in financial risk, insurance, and climate extremes. As with any generative model, outputs should be validated by domain experts before high-stakes use.

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A. Extreme Value Theory: Technical Details

This appendix provides the formal definitions and results from extreme value theory used in the main text. See (Resnick, 1987; de Haan & Ferreira, 2006) for a comprehensive treatment.

A.1. Heavy-Tailed Distributions

Definition A.1 (Heavy-Tailed (Nair et al., 2022))**.** A random variable X is *heavy-tailed* if $\mathbb{E}[e^{\lambda X}] = \infty$ for all $\lambda > 0$.

Definition A.2 (Regular Variation)**.** A measurable function $L : (0, \infty) \to (0, \infty)$ is *slowly varying* at infinity if $L(tx)/L(t) \to 1$ as $t \to \infty$ for all $x > 0$. A distribution F is *regularly varying* with index $-\alpha$ (written $F \in RV_{-\alpha}$) if its survival function satisfies $\bar{F}(t) = t^{-\alpha}L(t)$ for some slowly varying L.

The Pareto(γ) distribution with $\bar{F}(t) = t^{-1/\gamma}$ is regularly varying with index $-1/\gamma$ (i.e. $\alpha = 1/\gamma$); here $\gamma > 0$ is the shape parameter (extreme-value index).

A.2. The Fisher-Tippett-Gnedenko Theorem

Theorem A.3 (Fisher-Tippett-Gnedenko)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with distribution F. If there exist normalizing sequences $a_n > 0$ and $b_n \in \mathbb{R}$ such that*

$$\mathbb{P}\left(\frac{\max_{i \le n} X_i - b_n}{a_n} \le x\right) \to G(x)$$

for some non-degenerate distribution G, then G must be a generalized extreme value *(GEV) distribution:*

$$G_\xi(x) = \exp\left(-(1 + \xi x)^{-1/\xi}\right), \quad 1 + \xi x > 0,$$

where $\xi \in \mathbb{R}$ is the shape parameter *(extreme value index).*

The three cases correspond to:

- **Fréchet** ($\gamma > 0$): F has heavy (polynomial) tails; $F \in RV_{-1/\gamma}$
- **Gumbel** ($\gamma = 0$): F has light or sub-exponential tails (e.g. exponential, lognormal)
- **Weibull** ($\gamma < 0$): F has finite right endpoint (bounded support). Not to be confused with the Weibull *distribution* $W(k, \lambda)$ used in reliability analysis, which has unbounded support and lies in the Gumbel domain; the name clash is historical.

Proposition A.4 (Log-transform maps domains)**.** *If X is in the Fréchet domain with shape parameter $\gamma > 0$, then* $\log X$ *is in the Gumbel domain ($\gamma = 0$).*

This is the qualitative content of Proposition 3.8 and its precise restatement Proposition C.1: regularly varying tails $\bar{F}(t) = t^{-\alpha}L(t)$ become exponential-type after a logarithm, $\mathbb{P}(\log X > z) = e^{-\alpha z}L(e^z)$, and distributions with exponential-type tails are in the Gumbel domain (Embrechts et al., 2013, §3.3).

B. Denoising Diffusion Models: Technical Details

This appendix presents the framework underlying both denoising diffusion models and flow matching, organized around the *stochastic-interpolant* formulation of Albergo & Vanden-Eijnden (2023) and Lipman et al. (2023). We then recall how the canonical SDE-based diffusions of Ho et al. (2020) and Song et al. (2021b) arise as the special case in which the interpolation marginals coincide with those of an Ornstein–Uhlenbeck-type forward SDE.

B.1. Forward Noising Process

The stochastic interpolant of Albergo & Vanden-Eijnden (2023); Lipman et al. (2023) defines the probability path $(p_t)_{t \in [0,1]}$ via the marginals $p_t = \mathrm{Law}(X_t)$, where

$$X_t = \alpha_t X_0 + \beta_t X_1, \qquad X_0 \sim p_0, \quad X_1 \sim \mathcal{N}(0, I_d). \tag{4}$$

Here X_0 and X_1 are independent, and $(\alpha_t)_{t\in[0,1]}$ and $(\beta_t)_{t\in[0,1]}$ are deterministic schedules such that α_t is non-increasing, β_t is non-decreasing, with boundary conditions $(\alpha_0, \beta_0) = (1, 0)$ and $(\alpha_1, \beta_1) = (0, 1)$. This explicit-interpolation viewpoint is not the formalism of Ho et al. (2020); Song et al. (2021b), who instead define the forward process implicitly as the solution to a stochastic differential equation; the two viewpoints are reconciled in Section B.6.

From (4), the conditional distribution of X_t given $X_0 = x_0$, denoted $q_{t|0}$, is Gaussian:

$$q_{t|0}(x_t \mid x_0) = \mathcal{N}(x_t; \alpha_t x_0, \beta_t^2 I_d). \tag{5}$$

Common schedules. Two standard choices are the *variance-preserving* (VP) schedule with $\alpha_t^2 + \beta_t^2 = 1$ (Ho et al., 2020), which ensures $\mathrm{Var}(X_t) = \mathrm{Var}(X_0)$ when X_0 has unit variance, and the *linear* (flow matching) schedule with $(\alpha_t, \beta_t) = (1-t, t)$ (Lipman et al., 2023), corresponding to straight-line interpolation between data and noise. Table 6 summarizes common schedule choices with their derivatives.

Table 6. Interpolation schedules for flow matching. All satisfy boundary conditions $(\alpha_0, \beta_0) = (1, 0)$ and $(\alpha_1, \beta_1) = (0, 1)$. The linear schedule corresponds to optimal transport; VP schedules preserve variance when data has unit variance; the quadratic schedule anneals tails more aggressively early in the process.

Schedule	α_t	β_t	$\dot{\alpha}_t$	$\dot{\beta}_t$	Properties
Linear	$1-t$	t	-1	1	OT-optimal
VP (trig.)	$\cos(\frac{\pi t}{2})$	$\sin(\frac{\pi t}{2})$	$-\frac{\pi}{2}\sin(\frac{\pi t}{2})$	$\frac{\pi}{2}\cos(\frac{\pi t}{2})$	Smooth endpoints
VP (poly.)	$\sqrt{1-t}$	$\sqrt{t}$	$-\frac{1}{2\sqrt{1-t}}$	$\frac{1}{2\sqrt{t}}$	Singular endpoints
Quadratic	$(1-t)^2$	$1-(1-t)^2$	$-2(1-t)$	$2(1-t)$	Fast early annealing

B.2. Network Parameterizations: Noise vs. Data Prediction

Given a noisy observation x_t, two natural conditional-mean targets exist:

$$\hat{x}_0(x_t, t) := \mathbb{E}[X_0 \mid X_t = x_t], \qquad \hat{x}_1(x_t, t) := \mathbb{E}[X_1 \mid X_t = x_t].$$

These correspond to the two standard parameterizations in the diffusion literature (Ho et al., 2020; Song et al., 2021b): (i) **Noise prediction** (often called the *denoiser*, or ϵ-prediction in DDPM notation): $\hat{x}_1$ estimates the standard-Gaussian endpoint X_1, which plays the role of the additive noise in (4). (ii) **Data prediction** (also called the x_0-prediction or x_0-parameterization): $\hat{x}_0$ estimates the clean data sample. The two parameterizations are equivalent in expressive power: given $x_t = \alpha_t X_0 + \beta_t X_1$ from the forward interpolation (4), they satisfy the deterministic identity

$$\hat{x}_1(x_t, t) = \frac{x_t - \alpha_t \hat{x}_0(x_t, t)}{\beta_t}, \qquad \hat{x}_0(x_t, t) = \frac{x_t - \beta_t \hat{x}_1(x_t, t)}{\alpha_t}, \tag{6}$$

so one can be converted into the other at inference time. A third parameterization, v-prediction $v(x_t, t) := \mathbb{E}[\dot{\alpha}_t X_0 + \dot{\beta}_t X_1 \mid X_t = x_t]$, is the velocity field used by flow matching and is also a linear combination of $\hat{x}_0$ and $\hat{x}_1$.

B.3. Score-Denoiser Relationship

The *score* of the noised distribution is $\nabla \log p_t(\cdot)$. For the Gaussian forward kernel (5), the score admits a closed-form expression in terms of the denoiser.

Proposition B.1 (Score-Denoiser Identity)**.** *Under standard regularity assumptions,*

$$\nabla_{x_t} \log p_t(x_t) = \mathbb{E}\left[\nabla_{x_t} \log q_{t|0}(x_t \mid X_0) \,\middle|\, X_t = x_t\right] = -\frac{\hat{x}_1(x_t, t)}{\beta_t}. \tag{7}$$

Proof. The first equality is Fisher's identity, obtained by exchanging expectation and gradient. For the second, note that $\nabla_{x_t} \log q_{t|0}(x_t \mid x_0) = (\alpha_t x_0 - x_t)/\beta_t^2$. Taking the conditional expectation given $X_t = x_t$:

$$\mathbb{E}\left[\frac{\alpha_t X_0 - x_t}{\beta_t^2} \,\middle|\, X_t = x_t\right] = \frac{\alpha_t \hat{x}_0(x_t, t) - x_t}{\beta_t^2} = -\frac{\hat{x}_1(x_t, t)}{\beta_t},$$

where the last equality uses (6). □

Thus, learning the denoiser $\hat{x}_1$ is equivalent to learning the score $\nabla \log p_t$.

B.4. Training Objectives

The denoiser can be trained by regressing either X_0 or X_1 from the noised sample $X_t = \alpha_t X_0 + \beta_t X_1$. The X_0-prediction loss reads:

$$\mathcal{L}_{X_0}(\theta) = \int_0^1 \mathbb{E}_{X_0\sim p_0, X_1\sim\mathcal{N}(0,I_d)} \left[\left\| \hat{x}_0^\theta(\alpha_t X_0 + \beta_t X_1, t) - X_0 \right\|^2 \right] \mathrm{d}t,$$

while the X_1-prediction loss is:

$$\mathcal{L}_{X_1}(\theta) = \int_0^1 \mathbb{E}_{X_0\sim p_0, X_1\sim\mathcal{N}(0,I_d)} \left[\left\| \hat{x}_1^\theta(\alpha_t X_0 + \beta_t X_1, t) - X_1 \right\|^2 \right] \mathrm{d}t.$$

Since $\hat{x}_1^\theta = -\beta_t s_\theta$ by (7), the X_1-prediction loss is equivalent to denoising score matching (Hyvärinen, 2005; Vincent, 2011).

In practice, the integral is approximated via Monte Carlo: sample $t \sim \mathrm{Unif}[0,1]$, $x_0 \sim p_0$, $x_1 \sim \mathcal{N}(0, I_d)$, form $x_t = \alpha_t x_0 + \beta_t x_1$, and regress either x_0 or x_1.

B.5. DDIM Sampling

The DDIM framework (Song et al., 2021a) is canonically defined under the variance-preserving constraint $\alpha_t^2 + \beta_t^2 = 1$ and produces a one-parameter family of reverse transitions sharing the marginals of (5). Given timesteps $(t_k)_{k=0}^K$ with $t_K = 1$ and $t_0 = 0$, the transition from t_{k+1} to t_k is:

$$x_{t_k} = \alpha_{t_k}\, \hat{x}_0^\theta(x_{t_{k+1}}, t_{k+1}) + \sqrt{\beta_{t_k}^2 - \eta_{t_k}^2}\; \hat{x}_1^\theta(x_{t_{k+1}}, t_{k+1}) + \eta_{t_k} z, \tag{8}$$

where $z \sim \mathcal{N}(0, I_d)$ and $\eta_{t_k} \in [0, \beta_{t_k}]$ so that the radicand is nonnegative. Translating (Song et al., 2021a) eq. (12) into our notation (identifying their σ_t with η_{t_k}, $\sqrt{\alpha_{t-1}}$ with α_{t_k}, and $\sqrt{1-\alpha_{t-1}}$ with β_{t_k}) yields exactly (8). The family interpolates between two distinguished members: $\eta_{t_k} = 0$ gives the deterministic DDIM update, equivalent to a discretisation of the probability-flow ODE; the choice $\eta_{t_k} = \tilde{\beta}_{t_k}$, where

$$\tilde{\beta}_{t_k}^2 \;=\; \frac{\beta_{t_k}^2}{\beta_{t_{k+1}}^2} \Big(1 - \frac{\alpha_{t_{k+1}}^2}{\alpha_{t_k}^2}\Big), \tag{9}$$

is the posterior variance of $q(x_{t_k} \mid x_{t_{k+1}}, x_0)$ and recovers ancestral DDPM sampling (Ho et al., 2020; Song et al., 2021a): this matches Song et al. (2021a, eq. (16)) after the same change of variables. The boundary $\eta_{t_k} = \beta_{t_k}$ is the maximum-noise edge of the family, strictly noisier than DDPM.

B.6. Bridging to SDE-Based Diffusion Models

The interpolant formulation (4) is not the language used by Ho et al. (2020); Song et al. (2021b), whose forward processes are defined as solutions to discrete- or continuous-time stochastic differential equations. The two formalisms are equivalent at the marginal level for a specific schedule choice, and the equivalence is what Albergo & Vanden-Eijnden (2023) call the "unifying" aspect of stochastic interpolants. We make the bridge explicit in three steps: (i) recover Ho et al.'s DDPM forward chain; (ii) recover Song et al.'s forward score SDE; (iii) recover their reverse-time samplers.

(i) DDPM forward chain. Ho et al. (2020) define a discrete-time Markov chain

$$q(x_k \mid x_{k-1}) = \mathcal{N}\Big(x_k;\, \sqrt{1-\beta_k^{\mathrm{DDPM}}}\, x_{k-1},\, \beta_k^{\mathrm{DDPM}} I_d\Big), \quad k = 1, \dots, K, \tag{10}$$

with a noise schedule $(\beta_k^{\mathrm{DDPM}})_{k=1}^K \subset (0,1)$. Setting $\bar{\alpha}_k = \prod_{j\le k}(1-\beta_j^{\mathrm{DDPM}})$, marginalising the chain yields

$$q(x_k \mid x_0) = \mathcal{N}\big(x_k;\, \sqrt{\bar{\alpha}_k}\, x_0,\, (1-\bar{\alpha}_k) I_d\big)\,.$$

Identifying $t = k/K$, $\alpha_t = \sqrt{\bar{\alpha}_k}$, and $\beta_t = \sqrt{1-\bar{\alpha}_k}$ recovers (5) exactly. The DDPM noise schedule $(\beta_k^{\mathrm{DDPM}})$ is therefore one particular choice within the VP family in Table 6, with $\alpha_t^2 + \beta_t^2 = 1$ by construction.

(ii) Forward score SDE. Song et al. (2021b) unify diffusion models through the continuous-time Itô SDE

$$\mathrm{d}X_t = -\tfrac{1}{2}\beta(t)\,X_t\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}W_t, \qquad X_0 \sim p_0, \tag{11}$$

where $(W_t)_{t\geq 0}$ is a standard Brownian motion (the "VP-SDE"). The solution at time t has marginals

$$X_t \mid X_0 = x_0 \;\sim\; \mathcal{N}\Big(e^{-\frac{1}{2}\int_0^t \beta(s)\,\mathrm{d}s}\,x_0,\; \big(1 - e^{-\int_0^t \beta(s)\,\mathrm{d}s}\big) I_d\Big)\,,$$

so identifying $\alpha_t = \exp(-\frac{1}{2}\int_0^t \beta(s)\,\mathrm{d}s)$ and $\beta_t = \sqrt{1-\alpha_t^2}$ recovers (5). (11) is the continuous-time limit of the DDPM chain (10); both produce the same marginals as (4) under this matched schedule. Albergo & Vanden-Eijnden (2023) show that for any choice of (α_t, β_t) in the interpolant, there exists a (possibly time-inhomogeneous) SDE with the same marginals, so the interpolant is strictly more general than the VP family.

(iii) Reverse-time samplers. The marginal equivalence carries over to sampling. The time-reversed counterpart of (11) satisfies (Anderson, 1982):

$$\mathrm{d}\bar{X}_t = \big[-\tfrac{1}{2}\beta(t)\bar{X}_t + \beta(t)\nabla_x \log p_t(\bar{X}_t)\big]\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}\bar{W}_t, \tag{12}$$

with $\bar{X}_t := X_{1-t}$ and $\bar{W}_t$ a standard Brownian motion. The deterministic probability-flow ODE (Song et al., 2021b) with identical marginals reads

$$\mathrm{d}x_t = \big[-\tfrac{1}{2}\beta(t)x_t + \tfrac{1}{2}\beta(t)\nabla_x \log p_t(x_t)\big]\,\mathrm{d}t. \tag{13}$$

DDIM with $\eta_t = 0$ is a discretisation of (13); DDPM corresponds to (12). Flow matching learns a velocity field $v_\theta(x,t) \approx \mathbb{E}[\dot{\alpha}_t X_0 + \dot{\beta}_t X_1 \mid X_t = x]$ (Lipman et al., 2023); by (7), this velocity is a linear combination of identity and score, so any of the three samplers can be reconstructed from a single trained denoiser regardless of which formalism was used to derive the training loss.

C. Proofs

C.1. Precise statement and proof of Proposition 3.8

Proposition 3.8 of the main body states the rate-form conclusion $-\log \mathbb{P}(\tilde{X} > z) = \alpha z + o(z)$. We give here the corresponding precise two-sided Potter-bound statement and its proof. The result is classical in the regular variation literature; see Bingham et al. (1987, Chapter 1) for a comprehensive treatment.

Proposition C.1 (Log-Transform of Regularly Varying; precise)**.** *Let X be a nonnegative random variable that is regularly varying with index $-\alpha$, $\alpha > 0$. Set $\tilde{X} = \phi(X)$ with $\phi(x) = \operatorname{sign}(x)\log(1+|x|)$. Then for every $\epsilon > 0$ there exists $z_0 = z_0(\epsilon)$ such that for all $z \geq z_0$,*

$$e^{-(\alpha+\epsilon)z} \;\leq\; \mathbb{P}(\tilde{X} > z) \;\leq\; e^{-(\alpha-\epsilon)z}. \tag{14}$$

In particular, $-\log \mathbb{P}(\tilde{X} > z) = \alpha z + o(z)$ as $z \to \infty$.

Proof. By Karamata's representation, $\mathbb{P}(X > t) = t^{-\alpha}L(t)$ for some slowly varying function L. For the soft-log transform $\phi(x) = \operatorname{sign}(x)\cdot\log(1+|x|)$, we have $\phi^{-1}(z) = e^z - 1$ for $z \geq 0$, and $\phi^{-1}(z) \sim e^z$ as $z \to \infty$.

Thus

$$\mathbb{P}(\tilde{X} > z) \;=\; \mathbb{P}(X > e^z - 1) \;=\; (e^z-1)^{-\alpha}L(e^z - 1).$$

Since $e^z - 1 \sim e^z$, we have $(e^z-1)^{-\alpha} = e^{-\alpha z}(1+o(1))^{-\alpha}$ and, by slow variation, $L(e^z-1)/L(e^z) \to 1$. Hence $\mathbb{P}(\tilde{X} > z) = e^{-\alpha z}L(e^z)(1+o(1))$.

A standard consequence of slow variation (Resnick, 1987, Proposition 0.8(iv)) is that for every $\epsilon' > 0$, $L(t)/t^{\epsilon'} \to 0$ and $t^{\epsilon'}/L(t) \to 0$ as $t \to \infty$; equivalently, this follows from Potter's bounds (Bingham et al., 1987, Theorem 1.5.6) by fixing one argument at a large constant and absorbing the resulting multiplicative factor into the exponent. Either way, for any $\epsilon' > 0$ there exists $t_0 = t_0(\epsilon')$ such that for $t \geq t_0$,

$$t^{-\epsilon'} \;\leq\; L(t) \;\leq\; t^{\epsilon'}.$$

Substituting $t = e^z$ gives $e^{-\epsilon' z} \le L(e^z) \le e^{\epsilon' z}$ for $z \ge \log t_0$. Combining with $\mathbb{P}(\tilde{X} > z) = e^{-\alpha z} L(e^z)(1 + o(1))$ and absorbing the $1 + o(1)$ factor into a slightly enlarged ϵ (take $\epsilon' = \epsilon/2$ and z_0 large enough), we get

$$e^{-(\alpha+\epsilon)z} \;\le\; \mathbb{P}(\tilde{X} > z) \;\le\; e^{-(\alpha-\epsilon)z} \quad \text{for all } z \ge z_0(\epsilon),$$

which is (14). Taking $-\log$ and dividing by z yields $-\log \mathbb{P}(\tilde{X} > z)/z \to \alpha$, i.e. $-\log \mathbb{P}(\tilde{X} > z) = \alpha z + o(z)$. □

C.2. Proof of Proposition 3.9

Proof. Let $Y = X^\beta$. For the survival function of Y:

$$\bar{F}_Y(y) = P(X^\beta > y) = P(X > y^{1/\beta}) = \bar{F}_X(y^{1/\beta})$$

Then:

$$\begin{aligned} \frac{\bar{F}_Y(ty)}{\bar{F}_Y(t)} &= \frac{\bar{F}_X((ty)^{1/\beta})}{\bar{F}_X(t^{1/\beta})} \\ &= \frac{\bar{F}_X(t^{1/\beta} \cdot y^{1/\beta})}{\bar{F}_X(t^{1/\beta})} \\ &\to (y^{1/\beta})^{-\alpha} = y^{-\alpha/\beta} \end{aligned}$$

as $t \to \infty$, by the regular variation of X with index $-\alpha$. □

This extends Lemma 3.4: power transformation preserves the class of regularly varying distributions, with the tail index α scaling inversely with the exponent (equivalently, the shape parameter γ scaling linearly).

D. The Parametrized Soft-Log Family φ_{s_2}

The main body uses the soft-log $\phi(x) = \operatorname{sign}(x)\log(1+|x|)$ uniformly across coordinates, gated by a Hill mask (Section 4.2) when light-tailed margins are present. This appendix develops a continuous one-parameter generalization of ϕ that subsumes both the uniform ϕ and the binary mask as special cases, and clarifies its asymptotic behaviour. The construction is not used in our headline experiments; it is an extension we discuss as a research direction (Section 6).

Definition. For a scale $s_2 > 0$, define

$$\varphi_{s_2}(x) \;=\; \frac{1}{s_2}\,\phi(s_2\,x) \;=\; \frac{\operatorname{sign}(x)}{s_2}\,\log\bigl(1 + s_2\,|x|\bigr).$$

The family is parametrized so that $\varphi_1 = \phi$ (the soft-log of the main body) and $\varphi_{s_2}(x) \to x$ as $s_2 \to 0$ (the identity).

Bulk vs. tail. A Taylor expansion at $|x| = 0$ gives $\varphi_{s_2}(x) = x - \frac{1}{2}s_2 x|x| + O(s_2^2)$, so φ_{s_2} is the identity to leading order on the bulk $|x| \ll 1/s_2$. For $|x| \gg 1/s_2$, expanding $\log(1 + s_2|x|) = \log|x| + \log s_2 + \log(1 + 1/(s_2|x|))$ gives

$$\varphi_{s_2}(x) \;=\; \frac{\operatorname{sign}(x)}{s_2}\,\bigl(\log|x| + \log s_2\bigr) + O\bigl((s_2|x|)^{-1}\bigr),$$

which is logarithmic up to an additive constant and an overall $1/s_2$ scale. Thus s_2 acts as the inverse of the cross-over location: the bulk regime extends to $|x| \sim 1/s_2$, and the logarithmic-compression regime kicks in beyond. Choosing s_2 smaller makes the transform more conservative (closer to identity); choosing it larger compresses more aggressively.

Asymptotic tail-mapping. For any fixed $s_2 > 0$, Proposition 3.8 (and its precise restatement Proposition C.1) applies to $\tilde{X} = \varphi_{s_2}(X)$ unchanged up to a constant shift and rescaling: if $X \in RV_{-\alpha}$ then

$$-\log \mathbb{P}(\tilde{X} > z) \;=\; \alpha\, s_2\, z + o(z) \quad \text{as } z \to \infty.$$

The proof is identical to that of Proposition C.1 with $\phi^{-1}(z) = e^z - 1$ replaced by $\varphi_{s_2}^{-1}(z) = (e^{s_2 z} - 1)/s_2$. So φ_{s_2} still maps regular variation to exponential-type tails for any $s_2 > 0$; only the exponential rate is rescaled by s_2.

Coordinate-wise $s_2^{(j)}$ and the adaptive instance. In the multivariate setting we may pick a coordinate-dependent scale $s_2^{(j)}$. A simple data-driven choice is $s_2^{(j)} = c/\mathrm{IQR}(X_j)$ for a constant c (e.g. $c = 1$), which puts the cross-over at a robust scale of the marginal. The Hill-gated method of Section 4.2 is the discrete instance of this family with $s_2^{(j)} \in \{0, 1\}$, chosen by the diagnostic $\hat{\alpha}_j \le \alpha_{\max}$; coordinates flagged heavy-tailed get $s_2^{(j)} = 1$ (full soft-log) and the others $s_2^{(j)} = 0$ (identity). An outer scale $s_1^{(j)}$ rescales each $\tilde{X}_j$ to unit variance, which we apply during preprocessing in both the discrete and continuous variants.

Why the binary choice in the main body. Hill-type tail-index estimates are notoriously unstable as point estimates. A continuous $s_2^{(j)}$ derived from $\hat{\alpha}_j$ would inject this instability into the preprocessing pipeline; the binary rule uses $\hat{\alpha}_j$ only categorically (above or below $\alpha_{\max}$), so small perturbations leave the transform unchanged. This matches the stability principle of Section 3.1.

E. Experimental Details

This appendix collects the original Hickling Student-t benchmark used at submission time (Section E.1), the real-data Fama–French validation (Section E.2), the NLL validation table against Hickling & Prangle (2025), and the Log-FM hyperparameter listing. The headline benchmark is in Section 5; the experiments below complement it with the original setup the reviewers reference, and are kept for continuity.

E.1. Student-t Benchmark

We adopt the synthetic benchmark of Hickling & Prangle (2025, Section 4.1): $X_1, \ldots, X_{d-1} \overset{\text{iid}}{\sim} \text{Student-}t(\nu)$ and $X_d \mid X_{d-1} \sim \mathcal{N}(X_{d-1}, 1)$. We vary $d \in \{10, 20, 50\}$ and $\nu \in \{0.5, 1, 1.5, 2, 3, 5, 30\}$ (Table 7 reports $\nu \in \{1.5, 2, 3, 5\}$; the boundary regimes $\nu \in \{0.5, 1, 30\}$ are omitted for space). Each configuration uses 5,000 total samples (40/20/40 train/val/test split), averaged over 20 replications. Log-FM achieves the lowest W_1 across all configurations.

Table 7. Wasserstein-1 distance on the original Hickling benchmark, mean $\pm$ standard error over 20 replications. Bold = best.

d	ν	TTFfix	TTF	Log-FM
10	1.5	1.78 ± 0.24	17.3 ± 15.2	$\mathbf{0.63 \pm 0.02}$
10	2	0.63 ± 0.10	1.02 ± 0.31	$\mathbf{0.25 \pm 0.00}$
10	3	0.20 ± 0.01	0.90 ± 0.33	$\mathbf{0.15 \pm 0.00}$
10	5	$\mathbf{0.12 \pm 0.00}$	0.67 ± 0.20	0.14 ± 0.00
20	2	0.48 ± 0.03	6.01 ± 4.96	$\mathbf{0.23 \pm 0.00}$
20	3	0.20 ± 0.01	0.95 ± 0.24	$\mathbf{0.15 \pm 0.00}$
20	5	$\mathbf{0.13 \pm 0.00}$	1.80 ± 1.34	0.15 ± 0.00
50	2	0.74 ± 0.04	70.4 ± 64.6	$\mathbf{0.28 \pm 0.01}$
50	3	0.29 ± 0.01	2.73 ± 1.08	$\mathbf{0.22 \pm 0.01}$
50	5	$\mathbf{0.17 \pm 0.00}$	14.0 ± 12.7	0.19 ± 0.01

E.2. Real Financial Data

We evaluate on the Fama–French 5 industry portfolios ($d = 5$, daily returns 1963–2023). Hill estimation yields $\hat{\alpha} \approx 3.7$ across dimensions, placing the data in the intermediate-tail regime. Table 8 reports W_1 over 10 replications. gTAF achieves a marginal edge; Log-FM is competitive, and substantially better than TTF/TTFfix; mTAF fails catastrophically. On the high-dimensional S&P500 dataset ($d = 275$) all methods exhibit poor sample quality, dominated by architectural rather than tail-modelling limitations; we omit it.

E.3. ODE Solver Step Ablation

Table 9 sweeps the Euler step count K at sampling time. Quality plateaus by $K = 20$ and is essentially flat to $K = 500$.

E.4. Baseline Validation

To ensure fair comparison, we validate that our implementations of the baseline methods reproduce the results reported in Hickling & Prangle (2025). Table 10 compares our NLL values (per dimension) with their Table 7 reference values across

Table 8. W_1 on Fama–French 5 (mean $\pm$ std, 10 reps). gTAF marginally best; Log-FM competitive; mTAF diverges.

Method	W_1
gTAF	$\mathbf{0.127 \pm 0.007}$
Log-FM	0.133 ± 0.013
TTFfix	0.449 ± 0.069
TTF	0.487 ± 0.054
mTAF	5.0×10^4

Table 9. Solver ablation: W_1^P as a function of Euler step count. Gumbel, $d = 20$, $\tau = 0.5$, 10 reps. Less than 5% variation between $K = 10$ and $K = 500$.

α	10	20	50	100	200	500
1.5	0.521	0.543	0.568	0.579	0.584	0.588
2.0	0.137	0.135	0.138	0.141	0.142	0.143
2.5	0.084	0.081	0.083	0.084	0.085	0.086

the original benchmark configurations.

Table 10. NLL Validation: Our Implementation vs Hickling Reference (Table 7). Format: ours [ref]. The coupling-layer baselines (TTFfix, TTF) reproduce the reference values within ± 0.05 across all configurations; mTAF and gTAF reproduce the reference values closely in moderate-tail regimes ($\nu \geq 1$) but deviate substantially in pathological regimes ($\nu = 0.5$, especially $d = 50$), reflecting the training instabilities of these architectures reported in the original paper.

d	ν	TTFfix	TTF	mTAF	gTAF
5	0.5	3.32 [3.33]	3.32 [3.33]	4.05 [4.08]	5.70 [6.42]
5	1.0	2.34 [2.35]	2.34 [2.34]	2.53 [2.49]	2.53 [2.49]
5	2.0	1.89 [1.89]	1.88 [1.89]	1.91 [1.92]	1.90 [1.90]
5	30	1.47 [1.47]	1.47 [1.47]	1.46 [1.46]	1.47 [1.46]
10	0.5	3.55 [3.54]	3.54 [3.55]	4.59 [4.48]	7.86 [7.13]
10	1.0	2.48 [2.46]	2.47 [2.47]	2.66 [2.63]	2.68 [2.63]
10	2.0	1.94 [1.93]	1.94 [1.93]	1.96 [1.95]	1.95 [1.95]
10	30	1.48 [1.47]	1.48 [1.47]	1.47 [1.46]	1.48 [1.47]
50	0.5	3.71 [3.68]	3.71 [3.68]	6.43 [5.22]	21.44 [7.49]
50	1.0	2.58 [2.54]	2.58 [2.54]	3.14 [2.62]	3.66 [2.65]
50	2.0	2.01 [1.98]	2.01 [1.98]	2.04 [1.98]	2.06 [1.99]
50	30	1.52 [1.47]	1.52 [1.47]	1.50 [1.47]	1.51 [1.47]

TTFfix and TTF match within ± 0.03 across all configurations. mTAF and gTAF show larger deviations in pathological regimes ($\nu = 0.5$, $d = 50$), consistent with known instabilities reported in the original paper. This validates that our baseline implementations are faithful reproductions.

E.5. Hyperparameters

Table 11 summarizes all hyperparameters for Log-FM. Baseline hyperparameters follow Hickling & Prangle (2025).

Table 11. Log-FM hyperparameters.

Network	
Hidden dimension	256
Layers	4
Activation	SiLU
Time embedding	Sinusoidal (dim 256)
Training	
Optimizer	AdamW
Learning rate	5×10^{-3}
Weight decay	10^{-5}
Batch size	Full batch
Max epochs	5000
Early stopping patience	100
Gradient clipping	10.0
Sampling	
ODE solver	Euler
Integration steps	100
Output clamp	disabled ($c = \infty$; optional, inert for $\alpha \geq 1.5$)